ARITIFICIAL INTELLIGENCE(Q) – MAY 2026

KECERDASAN TIRUAN(Q) – MEI 2026

Generative Modeling

Mohammad Iqbal, S.Si., M.Si., Ph. D.

OUTLINES

- Introduction
- *Generative Modelling*
- *Latent Variable*
- *Autoencoder*
- *Variational Autoencoder*

Introduction

How do you think?

- 🙉Image
- 🙉Sound
- 🙉Text

ALL are FAKE!

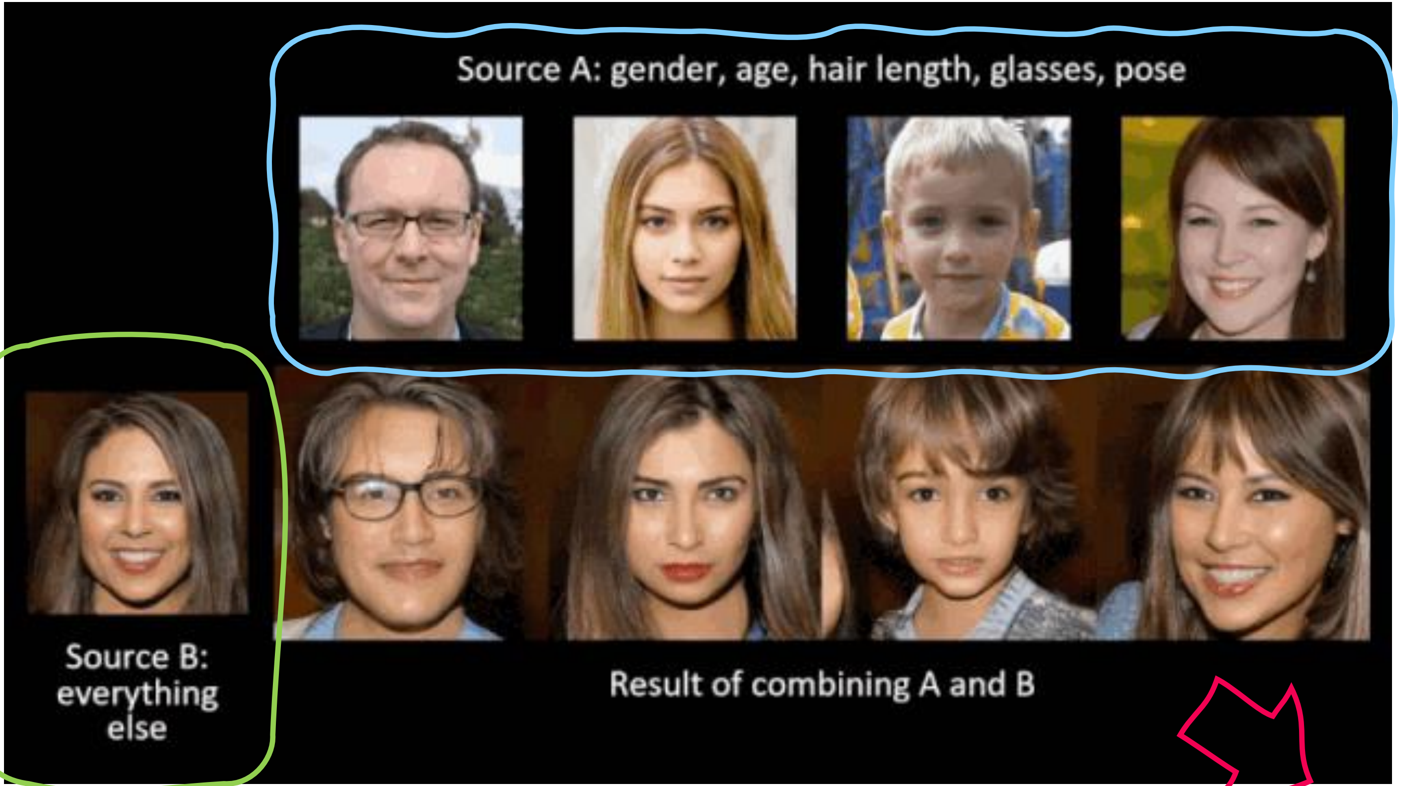
The idea : generation!

https://syncedreview.com/2019/02/09/nvidia-open-sources-hyper-realistic-face-generator-stylegan/

Deep FAKE

Gallery

Probe



FAKE!

The idea : generation! (cont'd)

https://www.youtube.com/watch?v=lME_giaXto8

SORA

The idea : generation! (cont'd)

https://www.youtube.com/shorts/jl38lYNBKpY

Prompt learning is the key!

Supervised Vs Unsupervised Learning

Supervised learning

Data : $\{ (x_i, y_i) \mid x, y \in \mathbb{R} \}$

$x \in X$: data, $y \in Y$: label/class

Goal : Learn function to map

$$\boldsymbol{f}: \boldsymbol{X} \rightarrow \boldsymbol{Y}$$

Example : classification, regression, object detection, semantic segmentation, etc.

Unsupervised Learning

Data : $\{ x_i \mid x \in \mathbb{R} \}$

$x \in X$: data (unlabeled)

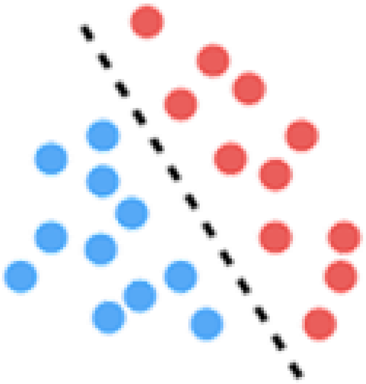
Goal : Learn the hidden or underlying structure of the data

Example : clustering, feature or dimensionality reduction, etc.

Generative vs Discriminative Models

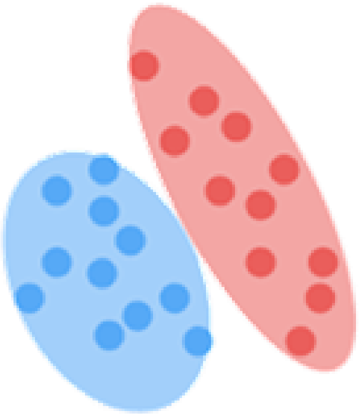
Discriminative model

- Estimate $p(y \mid x)$
- Supervised Learning Problem
- Decision boundary



Generative model

- Estimate $p(x, y)$:
  estimate $p(x \mid y)$ to obtain $p(y \mid x)$
- Unsupervised learning problem



Generative Modelling

Intuition **generative model**?

http://introtodeeplearning.com

Goals:

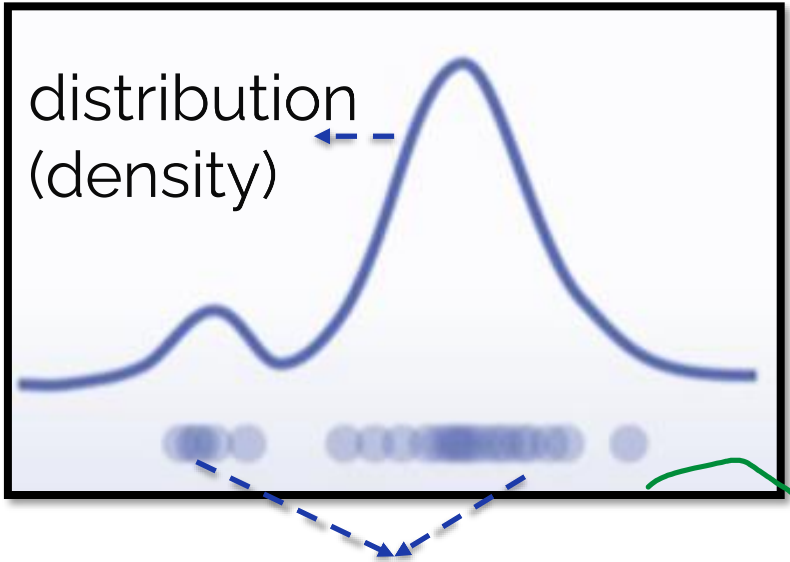
1. **Take** as input training samples from **some distribution**
2. **Learn a model** that **represents that distribution**

Our distribution looks like

http://introtodeeplearning.com

Density Estimation

To know our data distribution? **dense** or **sparse**.



Data (samples)

dense

0	12	2	0
7	3	8	0
0	0	42	1
9	7	0	6

sparse

	12		
7		8	0
		42	1
	7		

2D data ⇒ **3D** data?

Kernel Density Estimation

Kernel Density Estimation

https://www.youtube.com/watch?v=x5zLaWT5KPs&ab_channel=webelod

2D kernel

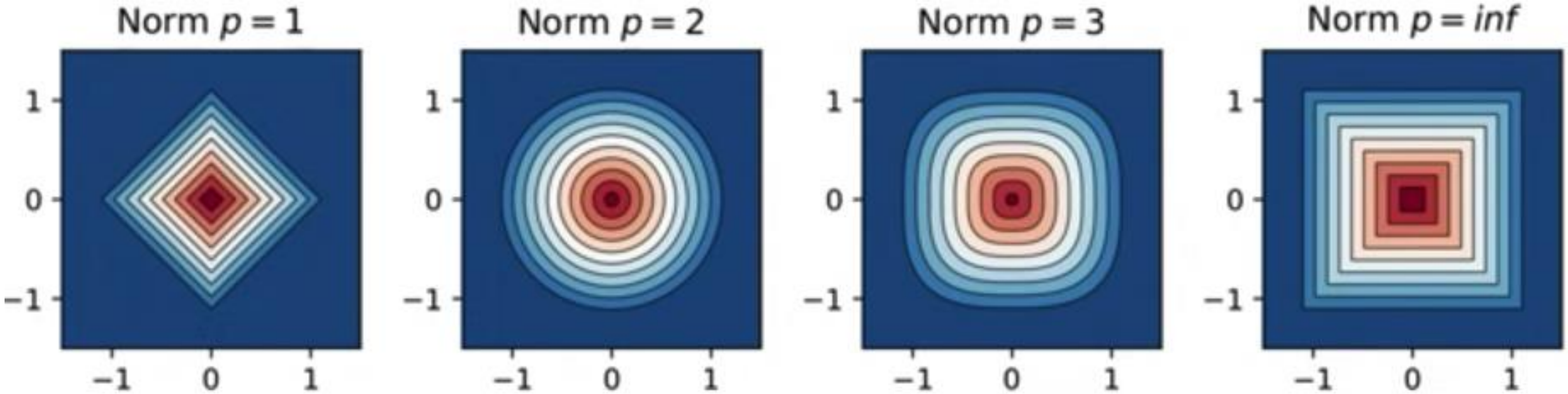
$$K(\|x - x_i\|)$$

Radial basis function

polynomial

p-norm

$$\|x\|_p := \left(\sum_i^n |x_i^p|\right)^{1/p}$$



Manhattan distance   Euclidean distance

SampleGeneration

Input Data

Training data $\sim P_{\text{data}}(x)$

Data Bangkitan

Generated data $\sim P_{\text{model}}(x)$

What is Generative Model

Learns $P_{\text{data}}(x)$ Such that **same as** $P_{\text{model}}(x)$

$G(x)$

Autoencoder

Variational Autoencoder

Generative Adversarial Networks

Why generative models?– Debiasing

http://introtodeeplearning.com/

Debiasing : reducing bias from consideration to decision making

VS

Capable for **uncovering underlying features in** dataset

How can we use this information to create fair and representative datasets?

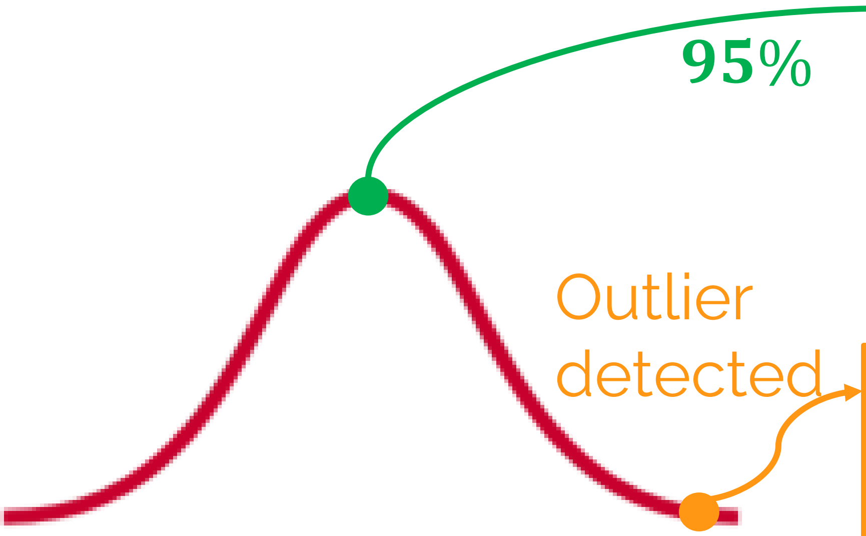
Why generative models? – Outlier Detection

http://introtodeeplearning.com/

Problem : How can we detect when we encounter somenting new or rare?

Strategy :

- Leverage generative models
- Detect outliers in the distribution
- Use outliers during training to improve even more!

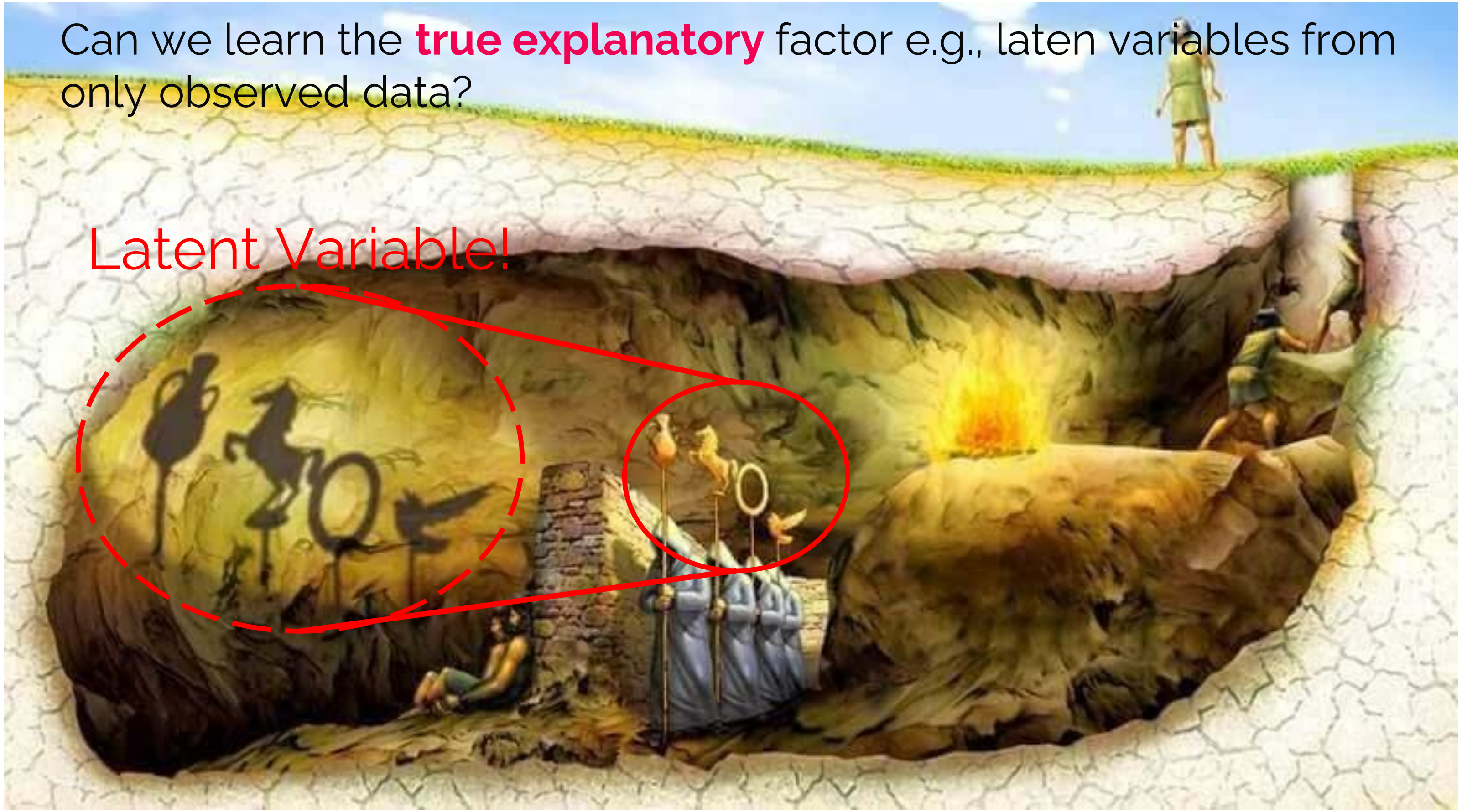


Sunny, highway, straight road

Latent Variable

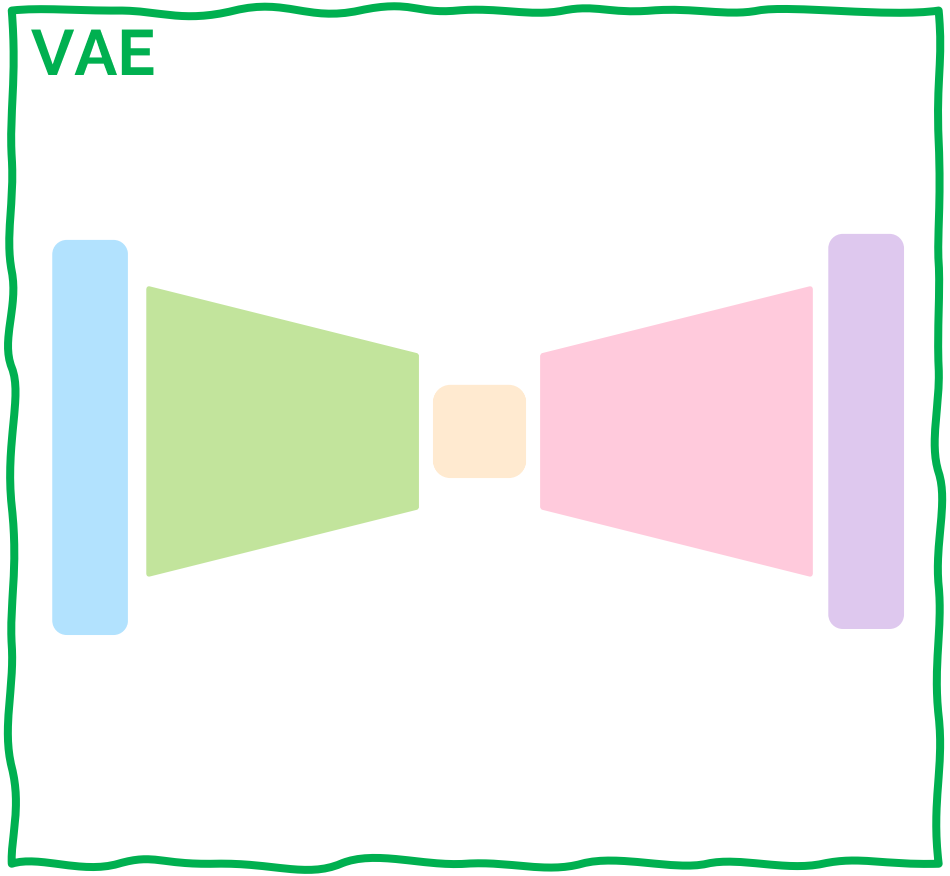
What is Latent Variable?

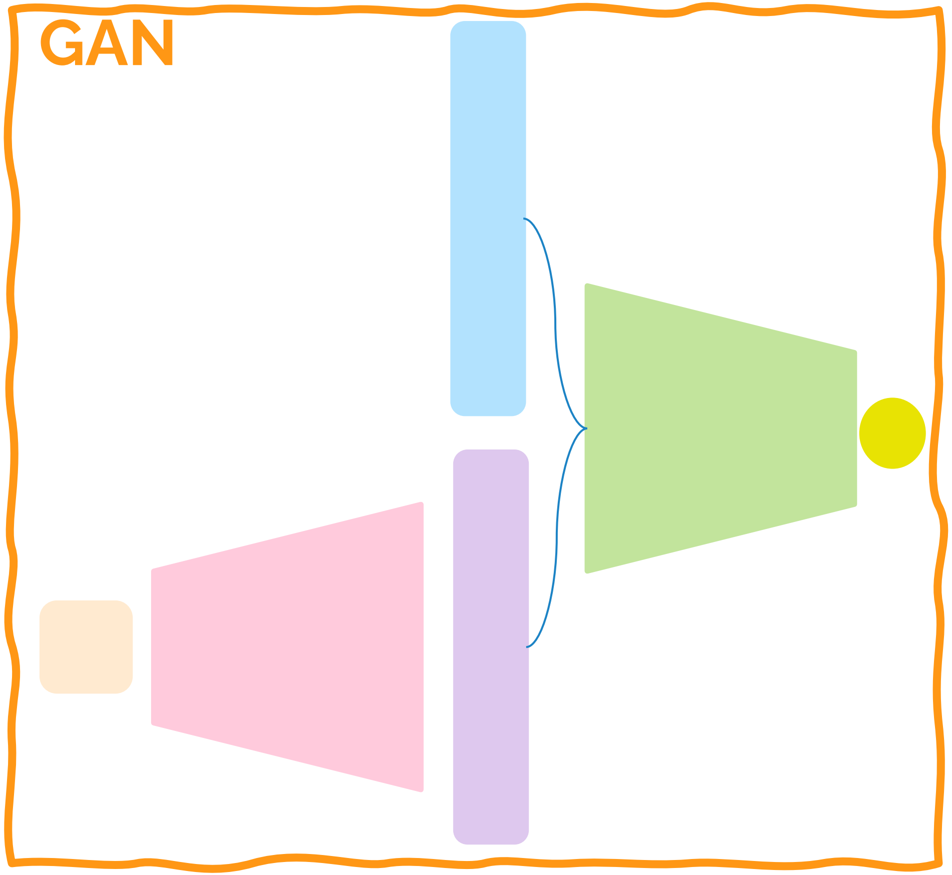
Can we learn the **true explanatory** factor e.g., laten variables from only observed data?



Plato Allegory—Myth Cave!

Latent Variable



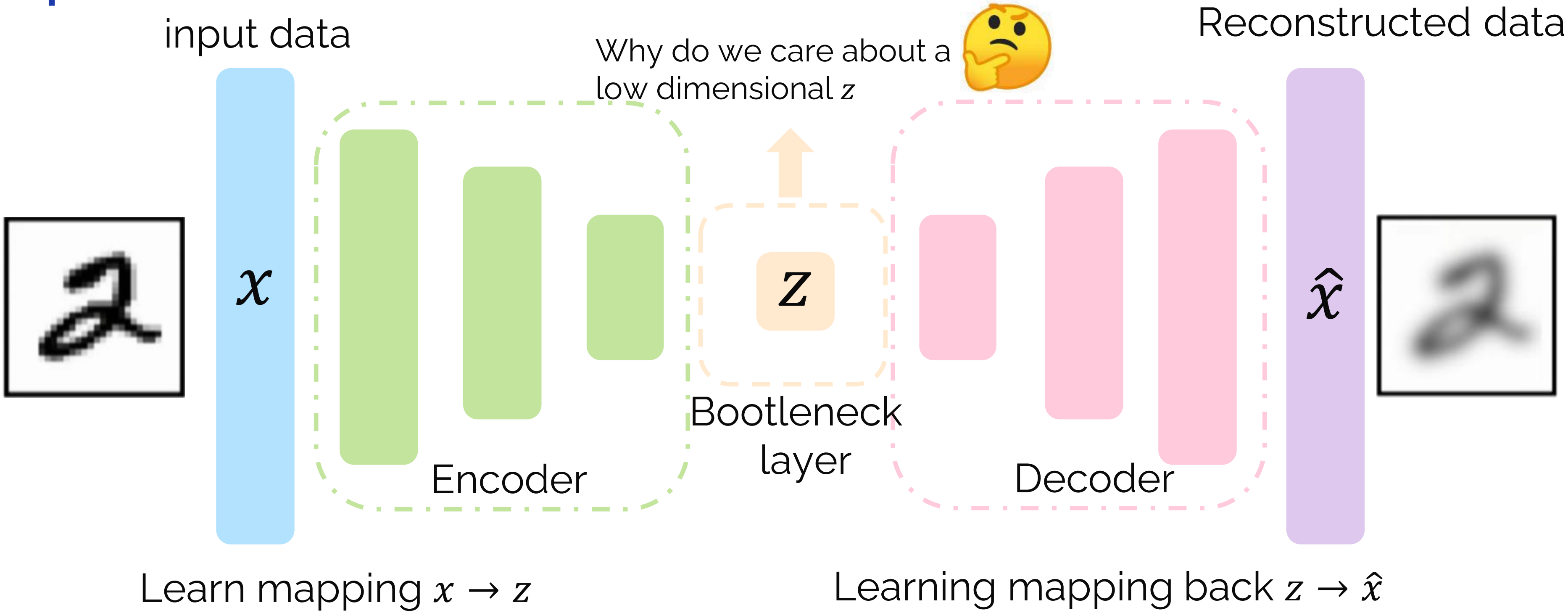


Autoencoders

Background: AE

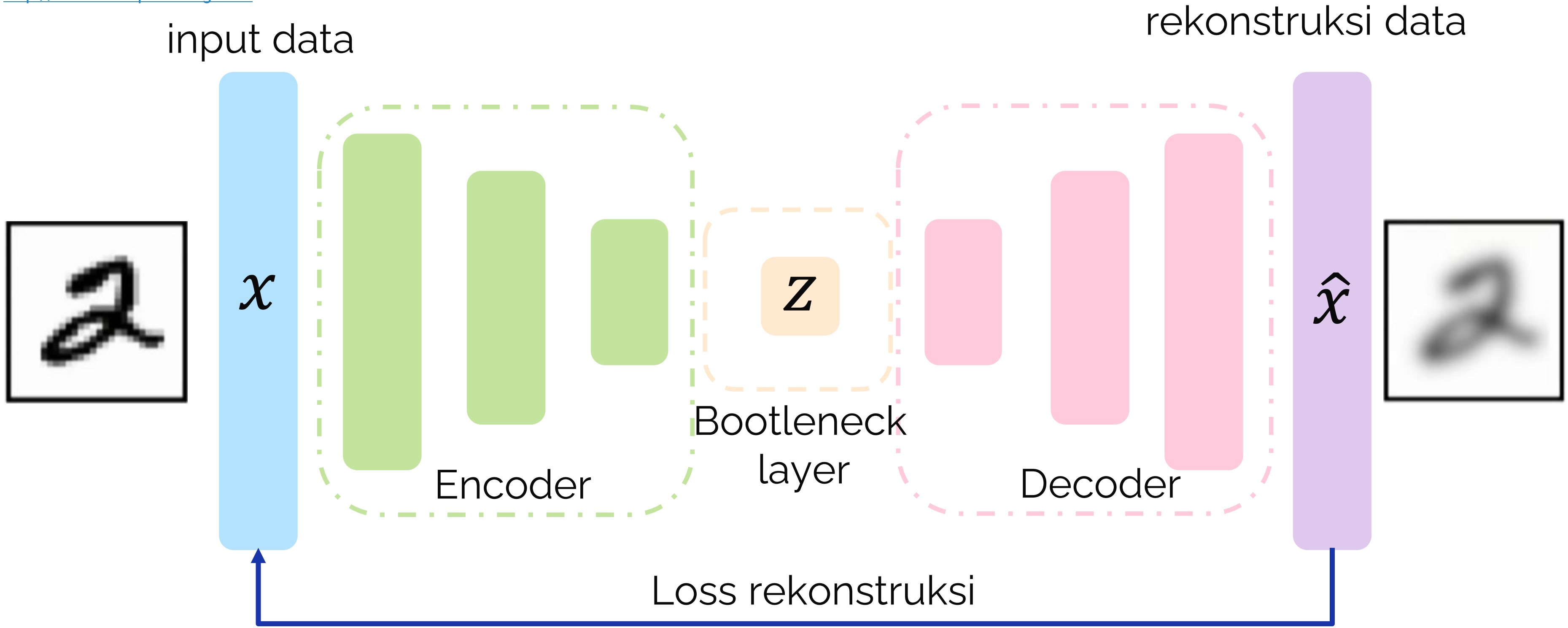
http://introtodeeplearning.com

Unsupervised approach for learning **a lower-dimensional feature representation** from **unlabeled data**



Try to enhance..

http://introtodeeplearning.com

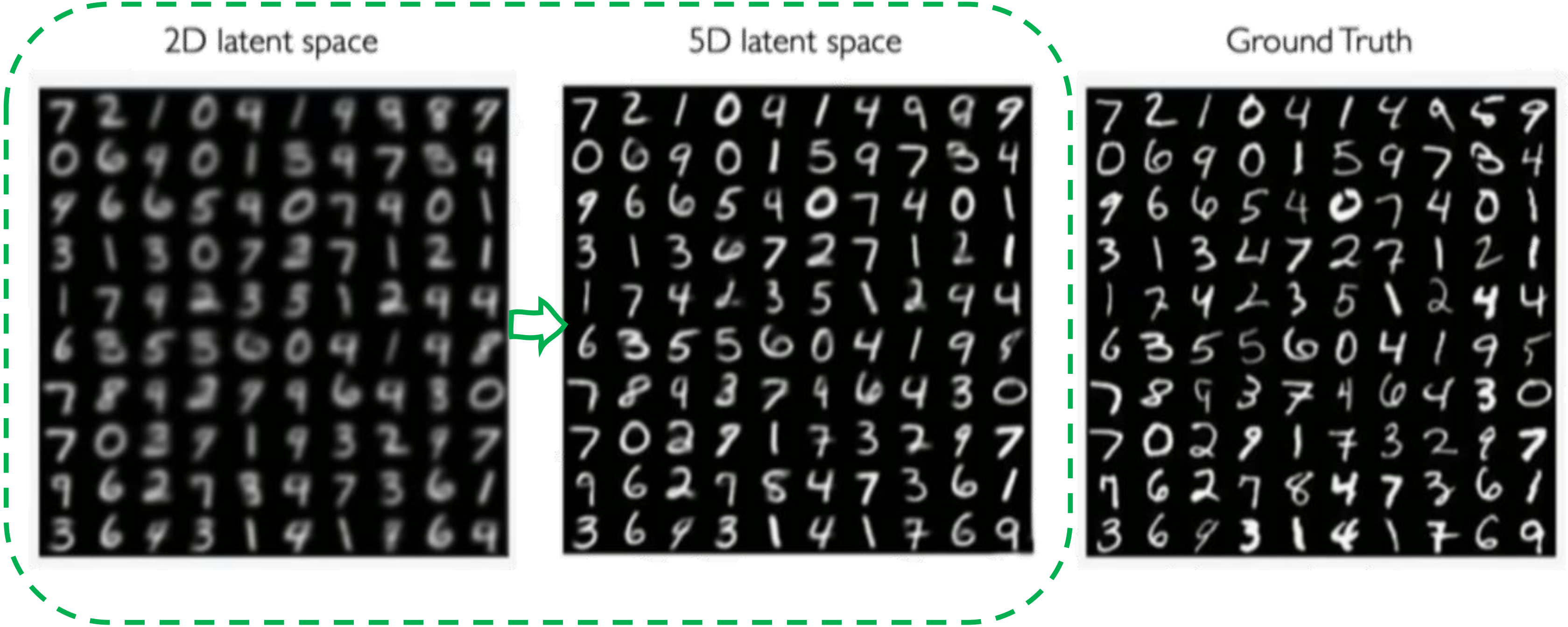


$$\mathcal{L}\,(x, \hat{x}) = \|x - \hat{x}\|^2$$

Latent Space Dimensionality~ Reconstruction quality

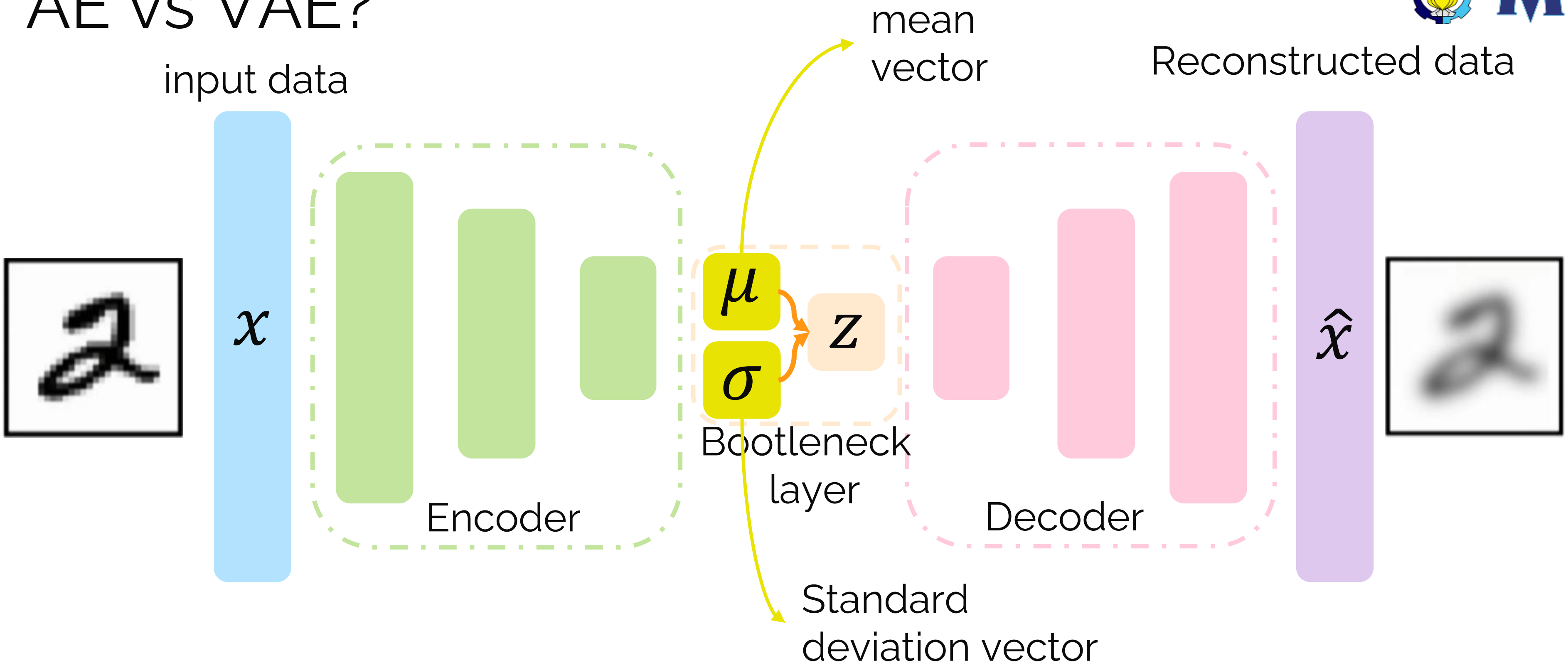
http://introtodeeplearning.com

- **Autoencoder** is a **form of compression!**
- **Smaller latent space will forcea larger training bottleneck**



Variational Autoencoders

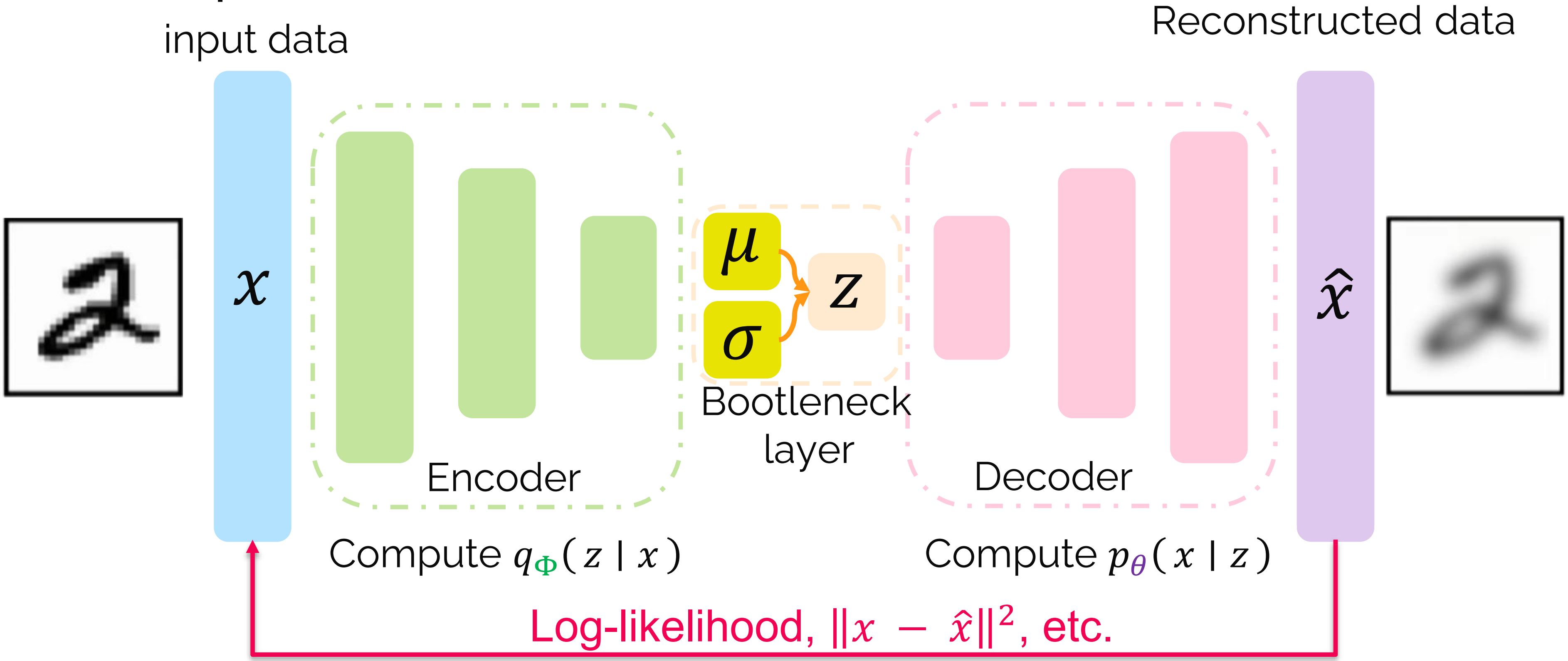
AE vs VAE?



VAEs are probabilistic twist on AE!

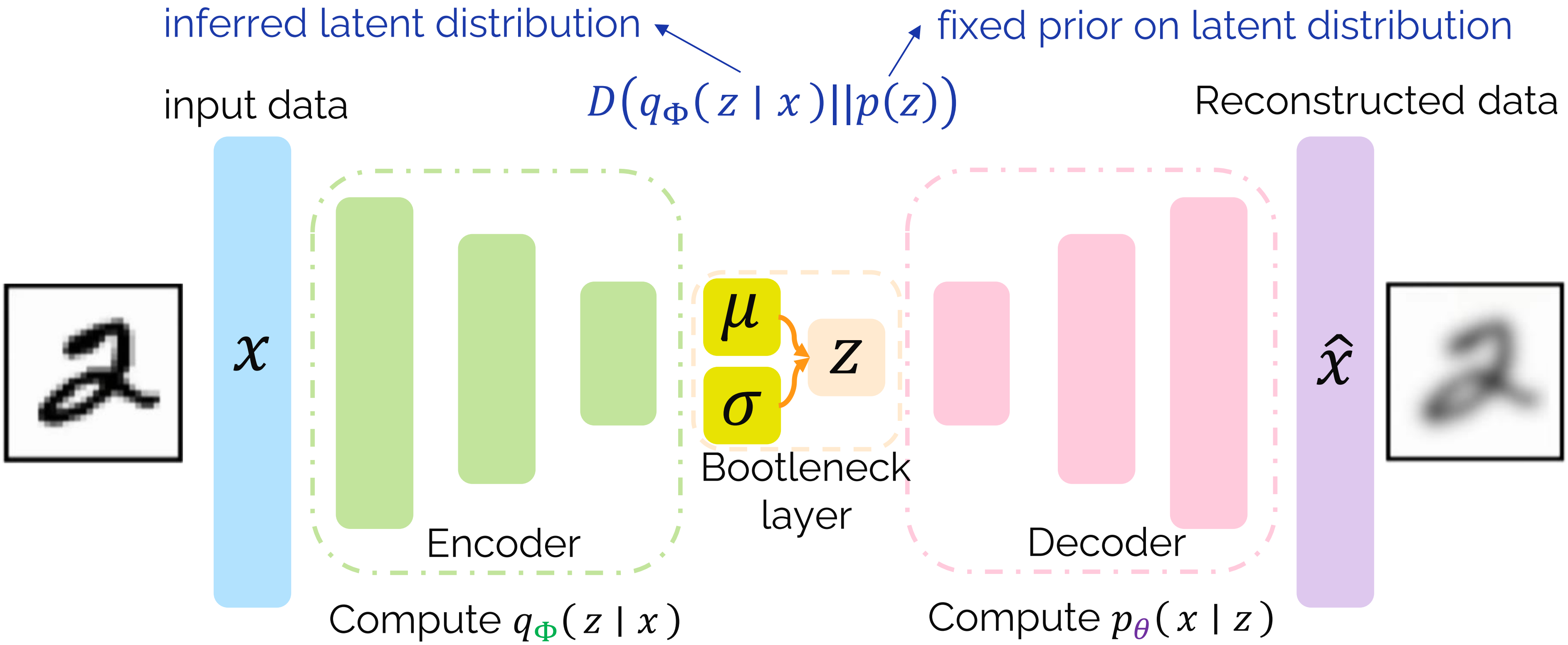
Sample from the mean and standard deviation to compute latent sample

VAE optimization



$\mathcal{L}(\Phi, \theta, x)$ = **reconstruction loss**

VAE optimization



$$\mathcal{L}(\Phi, \theta, x) = \textbf{recounstruction loss} + \textbf{regularization term}$$

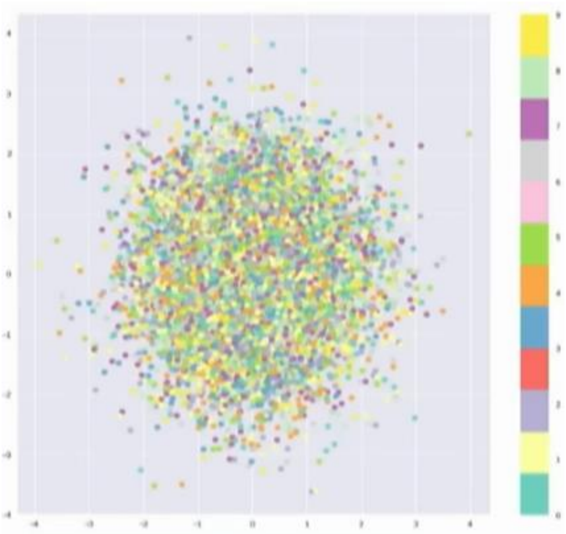
Priors on Latent Distribution

http://introtodeeplearning.com

$$D\left(q_{\Phi}(z \mid x) \| p(z)\right)$$

inferred latent distribution

fixed prior on latent distribution



Common choice of prior ~ Normal Gaussian:

$$p(z) = \mathcal{N}(\mu = 0, \sigma^2 = 1); z \sim \mathcal{N}(0,1)$$

- Encourage encodings to distributed encodings evenly around the center of the latent space.
- Penalize the networks when it tries "cheat" by clustering poits in specific regions (i.e.,memorizing the data)

Priors on Latent Distribution

$$D\left(q_{\Phi}(z \mid x)\|p(z)\right) = -\frac{1}{2}\sum_{j=0}^{k-1} \sigma_j + \mu_j^2 - 1 - \log \sigma_j$$

Kull-back Leiber Divergence between two distribution

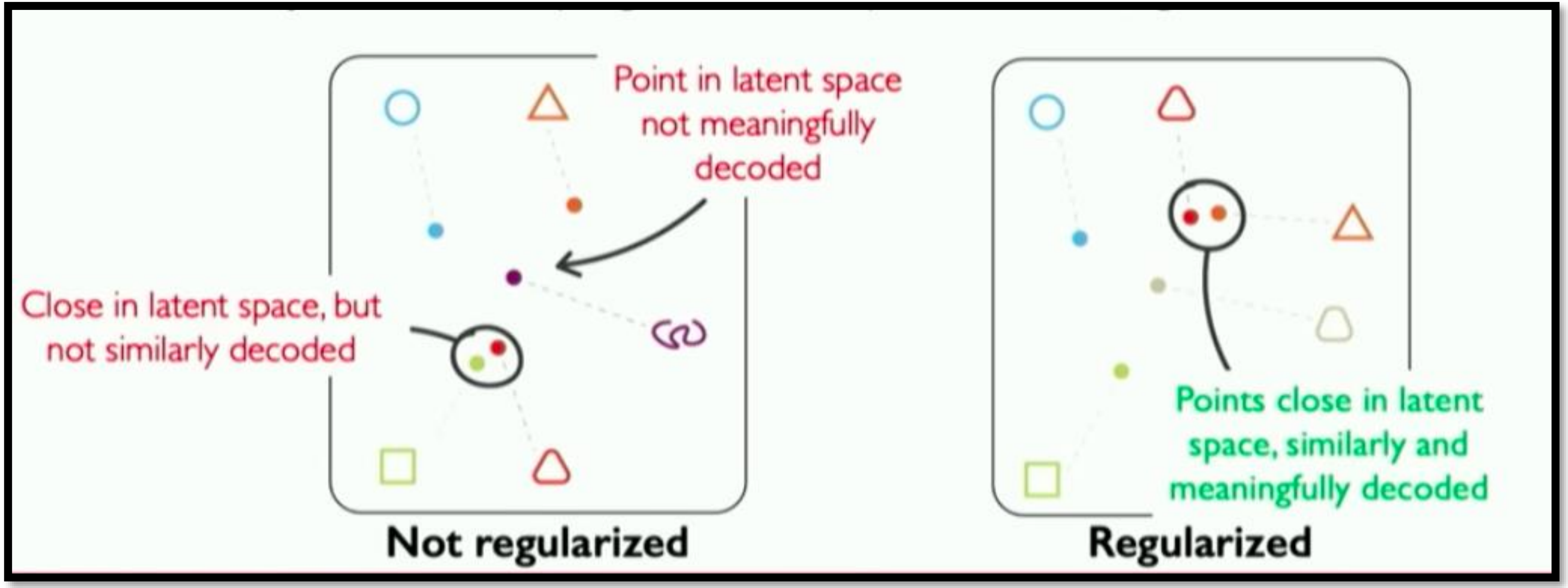
$$D(p(x) \,||\, q(x)) = \sum_{x \in X} p(x) \frac{p(x)}{q(x)}$$

Intuition: regularization and the normal prior

http://introtodeeplearning.com

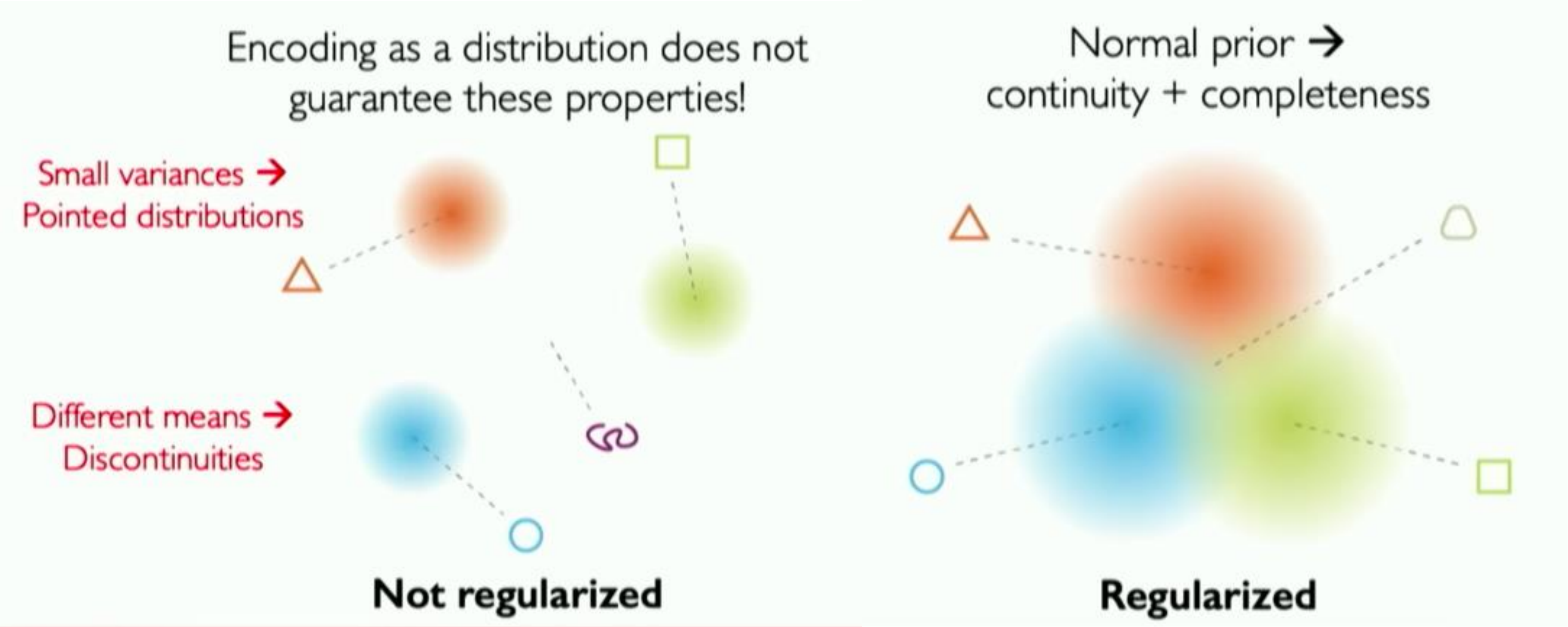
The goal of regularization:

- **Continuity** : points that are close in latent space→ **similar content after decoding**
- ***Completeness*** : sampling from latent space → **"meaningfull" content after decoding**



Intuisi: regularisasi dan normal prior

http://introtodeeplearning.com



Regularization with Normal prior helps to enforce **information gradient** **in** the latent space

Komputasi graph VAE

Problem : Cannot Backpropagate gradient through sampling layer!

Idea :
Reparameterizing sampling layer

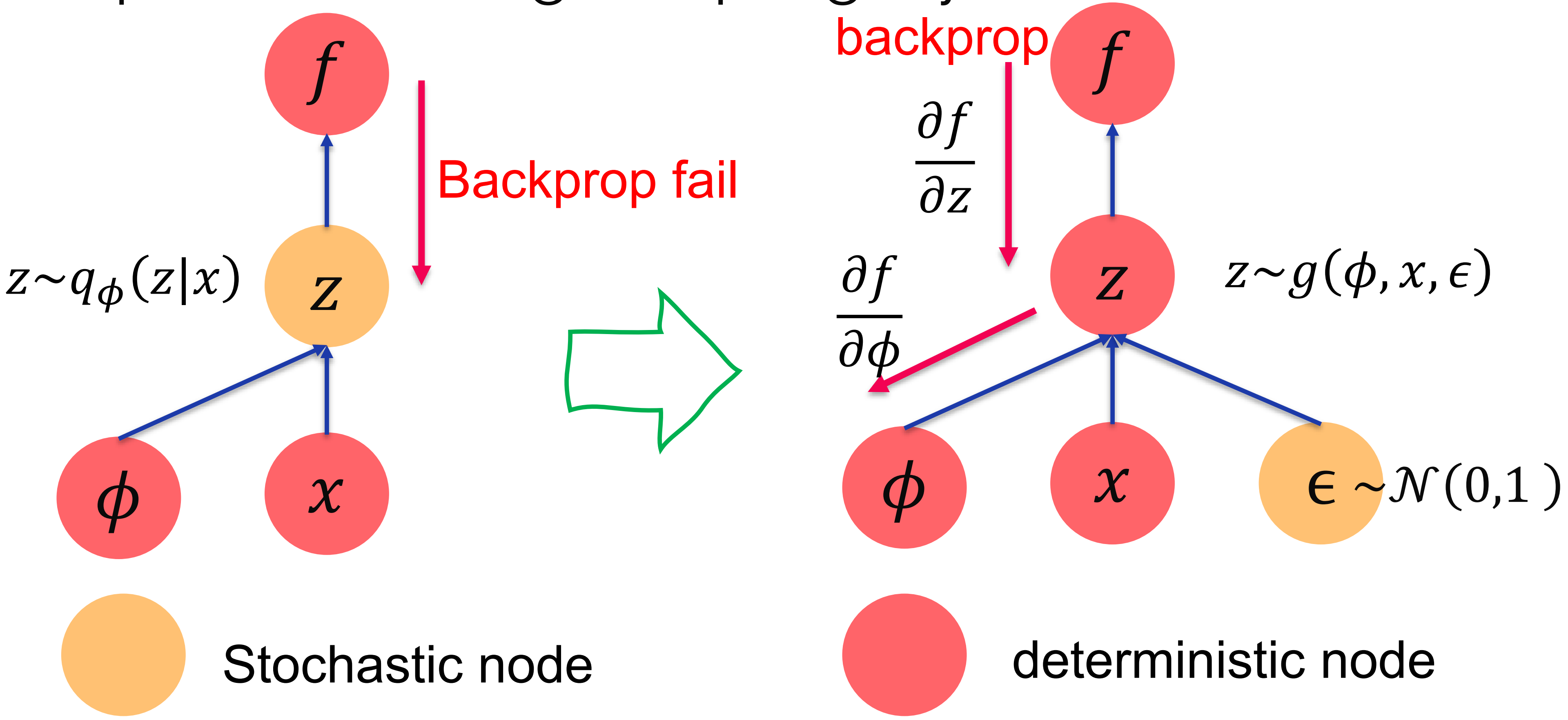
$$z \sim \mathcal{N}(\mu, \sigma^2)$$

$$\Downarrow$$

$$z = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \epsilon$$

σ scaled by random constants drawn from the prior distribution $\epsilon \sim \mathcal{N}(0,1)$

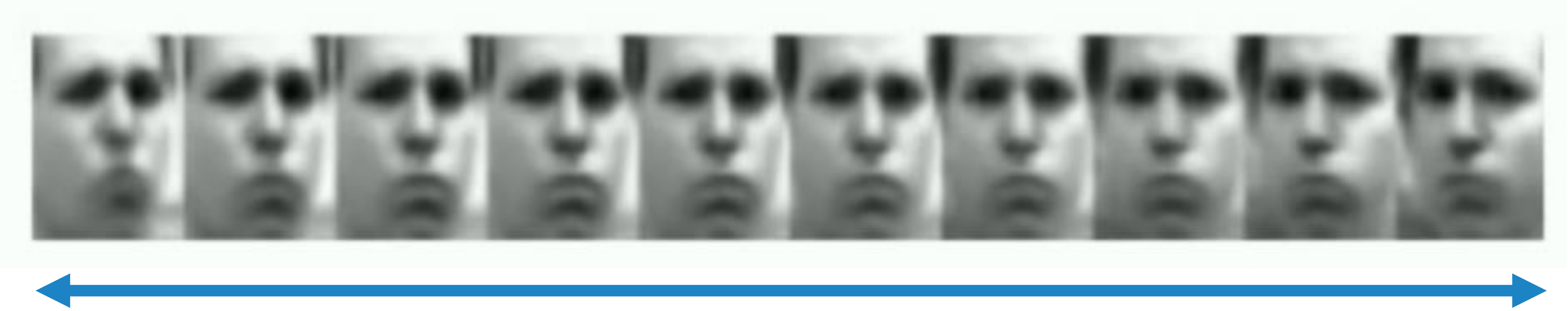
Reparameterizing sampling layer



Latent Peturbation: VAE

http://introtodeeplearning.com

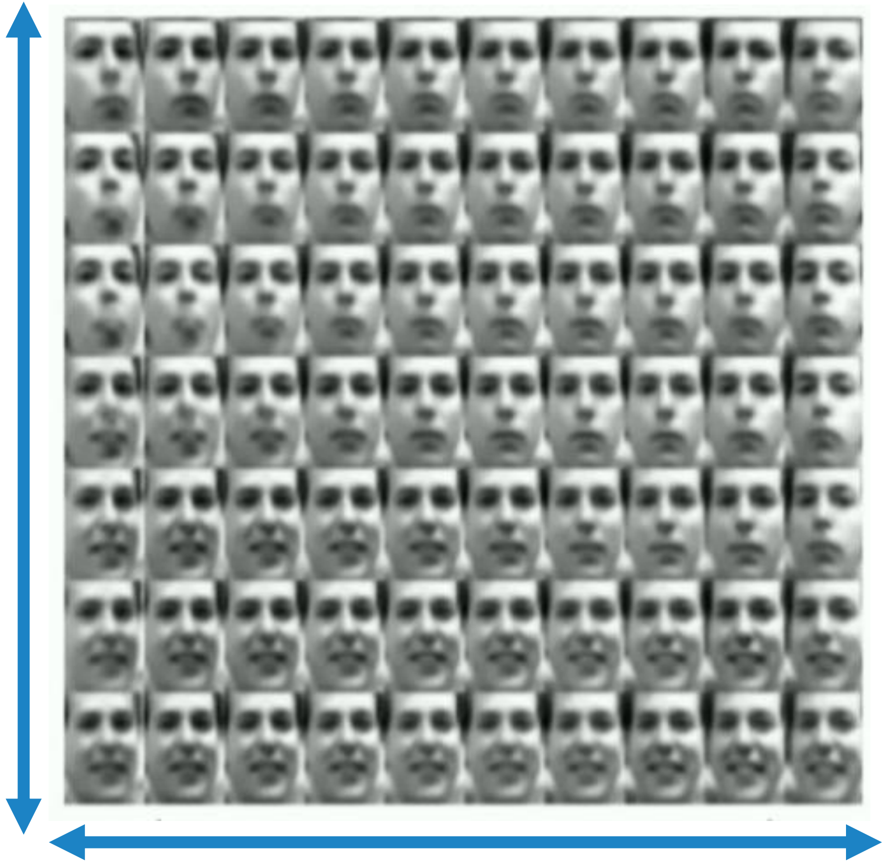
- Slowly increase or decrease a **single latent variable**.
- Keep all other variables fixed



- **Different dimensions of z** encodes **different interpretable latent features**

Latent Peturbation: VAE

http://introtodeeplearning.com



- Ideally, we want latent variables that are uncorrelated with each other.
- Enforce diagonal prior on the latent variables to encourage independence

Distanglements

Latent Space Distanglement β-VAE

http://introtodeeplearning.com

Traditional VAE:

$$\mathcal{L}(\phi,\theta;x,z,\beta) = \mathbb{E}_{q\_\phi(z|x)}[\log p_\theta(x|z)] - D_{KL}\left(q_\phi(z|x)||p(z)\right)$$

loss rekonstruksi **regularisasi term**

β VAE:

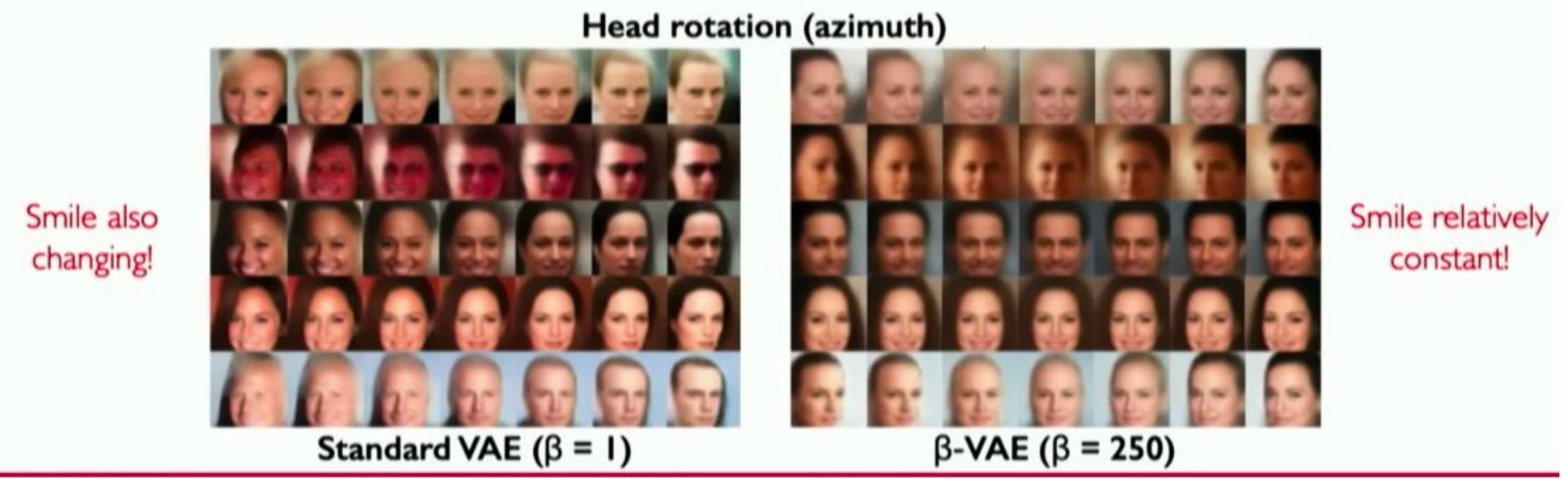
kontrol

$$\mathcal{L}(\phi,\theta;x,z,\beta) = \mathbb{E}_{q\_\phi(z|x)}[\log p_\theta(x|z)] - \beta\, D_{KL}\left(q_\phi(z|x)||p(z)\right)$$

$\beta > 1$: encourage efficient latent encoding → distanglement

Latent Space Distanglement β-VAE

http://introtodeeplearning.com



Why latent variable models?– Debiasing

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VS

Capable for **uncovering underlying features in** dataset
--> **Latent Variables**

Discussion...

Q n A